School of Computing and Mathematical Sciences

CO7201 Individual Project

Interim Report

Software Bug Tracking & Reporting Tool

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DECLARATION

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1. OVERVIEW

The Bug Tracker project aims to develop a robust and an open sourced, full-stack software bug tracking and reporting tool using modern web technologies. The system is being built with a React + TypeScript + Tailwind CSS + ShadCN frontend and a Spring Boot backend with PostgreSQL as the primary database. The objective is to deliver a practical, usable open-sourced platform to streamline issue tracking, team management, and project reporting for software development teams. This interim report summarizes progress to date, changes made to the initial plan, key challenges encountered and presents an updated project timeline.

2. TASK COMPLETION STATUS

The work has been structured in line with the original project plan, focusing on iterative, end-to-end delivery of critical modules:

Module/ Task	Start Date	End Date	Status	Details
User Authentication (DB, Backend, Frontend, Test)	08/06/2025	22/06/2025	Completed	Fully functional JWT-based registration and login system; UI validated and tested end-to-end.
Teams Module (DB, Backend, Frontend)	29/06/2025	05/07/2025	Completed	Users can create and join teams. Role management implemented.
Project Module (DB, Backend, Partial Frontend)	06/07/2025	12/07/2025	Partially Completed	Core backend APIs finished; basic frontend integration underway.
Bug Reporting Module	13/07/2025	19/07/2025	Pending	To be started.
UI Polish & MVP Testing	20/07/2025	25/07/2025	Pending	Planned post-feature implementation.
Duplicate Bug Detection (Prototype)	01/08/2025	08/08/2025	Pending	Planned advanced feature.
User Notification System	09/08/2025	15/08/2025	Pending	Planned advanced feature.
Data Export (CSV/Excel)	16/08/2025	22/08/2025	Pending	Planned advanced feature.

Final Documentation & Submission Prep	30/08/2025	05/09/2025	Pending	To include final documentation, diagrams, and university submission.

3. CHANGES TO INITIAL PLAN

The overall structure of the project remains unchanged; however, certain advanced features and deployment steps have been adjusted or de-prioritized based on feedback and scope management. Notably:

- **Deployment:** Deployment to live servers has been removed as it is not required.
- **Advanced Features:** Focus has narrowed to duplicate bug detection, notification, and export features as advanced enhancements. Other potential features have been dropped to maintain a realistic and high-quality submission.
- **Frontend/Backend:** Implementation is now strictly module-based, ensuring end-to-end functionality for each core area before moving to the next.

4. CHALLENGES

Several challenges have been encountered so far:

- **Balancing Scope and Timeline:** Prioritizing core features over optional enhancements to meet university deadlines and ensure a working MVP.
- **Frontend–Backend Sync:** Ensuring smooth integration between frontend components and backend APIs, particularly as requirements evolved.
- **Learning Curve:** Gaining proficiency with new frontend libraries (e.g., Tailwind CSS, advanced React hooks) and maintaining industry-standard practices for code quality, documentation, and version control.
- **Testing:** Manual testing has been performed for completed modules, but the setup for automated testing (especially for frontend) remains a future task.
- **Architectural Refactoring:** Having to rework the Teams module after realizing the Project module should have been implemented first delayed progress but provided important lessons in planning and design.

Module Sequencing Error and Refactoring

A significant challenge encountered so far was the sequence in which the core modules were implemented. Initially, I developed the Teams module before the Project module, even though both are tightly coupled and the Project module logically serves as the parent. This was a misstep in the architecture. Projects should contain teams, and not the other way around.

As a result, I am now engaged in refactoring the Teams module to properly align with the intended system design. This refactoring involves:

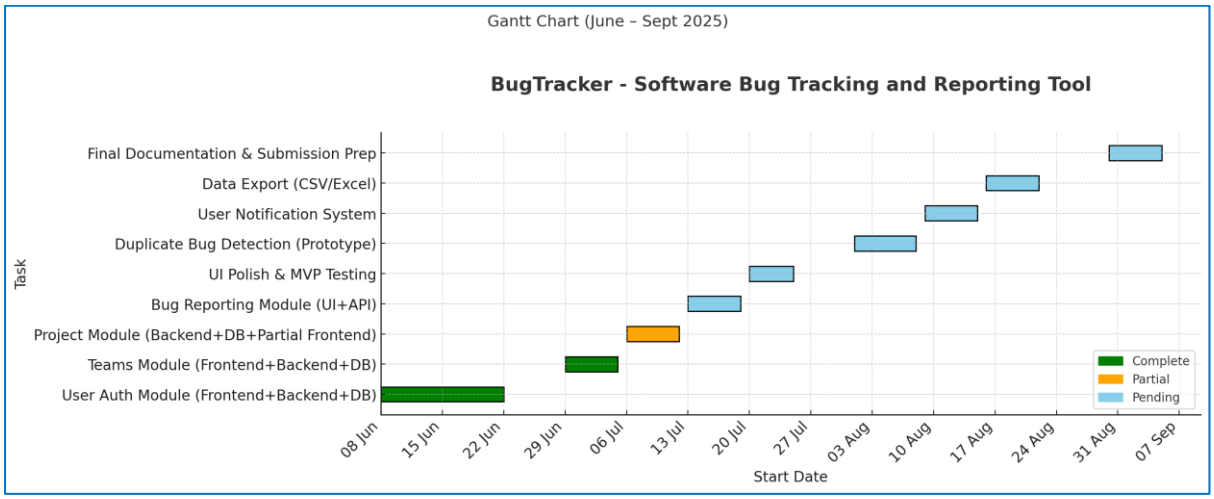
- Adjusting data models so that projects serve as containers for teams.
- Modifying API endpoints and relationships to reflect this hierarchy.

- Updating frontend logic to ensure seamless navigation and association between projects and their respective teams.

While this has temporarily slowed development, it is an important lesson in requirements analysis and software architecture, and will result in a cleaner, more maintainable system going forward.

5. UPDATED TIMELINE

The updated timeline is reflected in the Gantt chart below, highlighting completed, in-progress, and planned tasks:



6. REVISED PLAN

Module/ Task	Revised Plan	Status/ Notes
User Auth Module	8 – 22 June	Completed
Teams Module	29 June – 5 July	Implemented, refactoring now
Project Module	6 – 12 July	Refactoring in progress
Project Module Code Refactoring	24 July	Should be done by EOD
Teams/ Project Integration	25 July	Complete integration
End to End testing	26 – 27 July (Weekends)	Testing everything implemented so far
Bug Reporting Module	Start on 28 July	Planned
Data Export	1 – 3 August	Planned advanced feature
Duplicate Bug Detection	5 – 10 August	Planned advanced feature
User Notification System	11 – 20 August	Planned advanced feature
Testing phase	21 Aug – 30 Aug	Testing everything, bug fixes, improvements, polishing the application
Final documentation and submission prep	1 Sep onwards	Extensively work only on the documentations